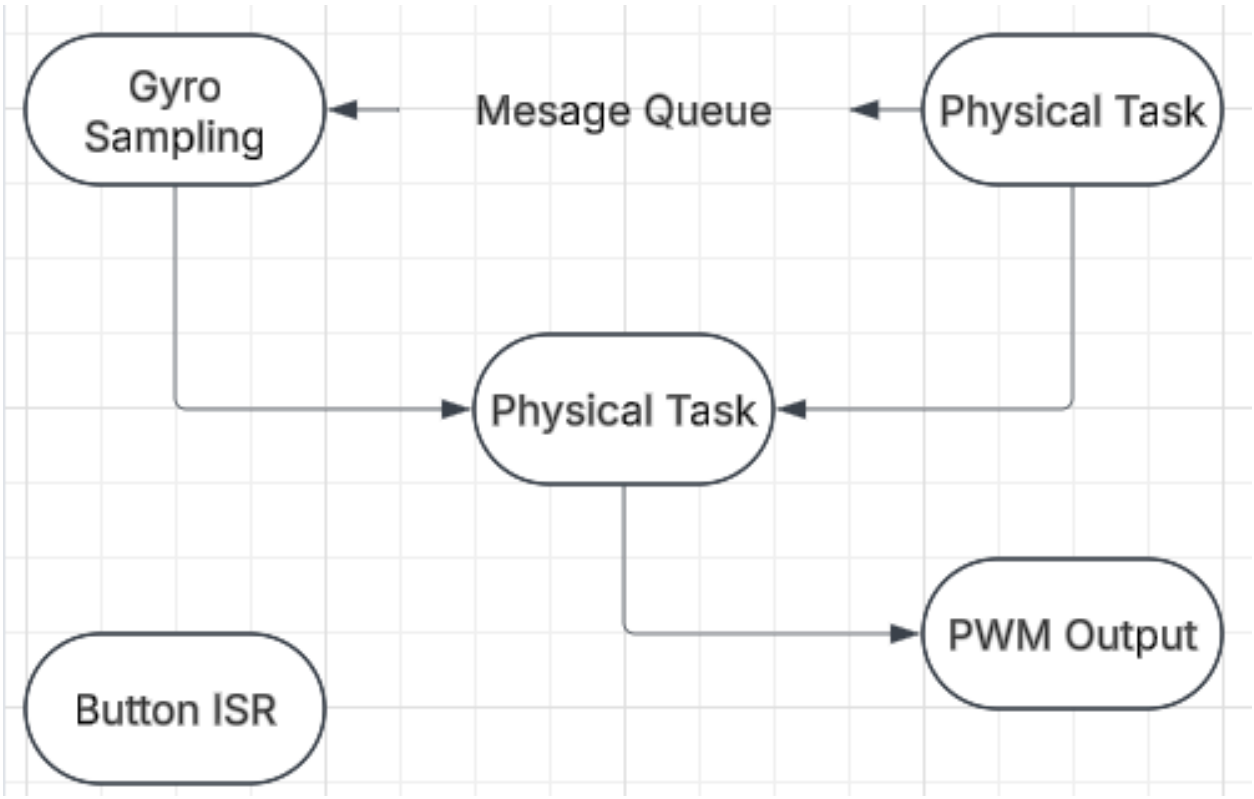
Joel J. Ramos Silva
ECEN 3753 RTOS

Step 1: Task Diagram



Step 2: Test Plan

- Button Press Handling
  - If the button press is being detected,
  - I want to simulate the short press and the long press.
  - Check the boundary condition
- Platform Collision Physics.
  - Platform collision detection and velocity logic
  - Give platform location and apply a unit of velocity
  - If the correct collision happens sends a message

Step 3: Status Summary

This week I read the project specification, created the initial task diagram with lucid chart using flows, and defined two appropriate unit tests for button logic and platform collision. I also planned my work breakdown and began organizing my GitHub structure for the embedded project.

Joel J. Ramos Silva
ECEN 3753 RTOS

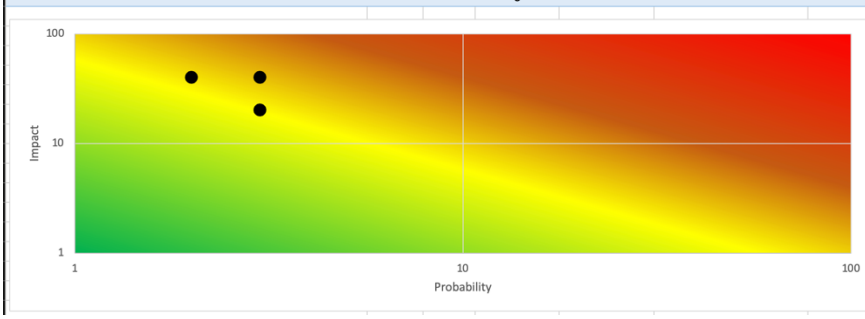
Summary effort & estimate numbers

I have completed 15% of my estimated work (6 estimated hours completed out of 40 total estimated) in 11% of the project time budgeted (5 hours spent so far out of 40). For the work completed, I took 2x as much

Step 4: Task Table

Task	Status	Est Time (Hrs)	Actual Time (Hrs)	Notes
Pysics Engine	Complete	2	4	Completed
Gyro	In progress	3	5	Stay Refining the numbers
LCD Refresh	Complete	2	3	Done

This week, I worked on the physics engine, gyro angle integration, and LCD display update. The physics engine took longer than expected because I needed to review mechanics, especially for elastic collisions and force-based speed updates, and to debug early velocity issues tied to the display. For the gyro angle, I implemented a rolling angle calculation from velocity, which went smoothly thanks to prior lab. The LCD update also took extra time as I had to figure out the screen coordinate system and scale everything to match real-world units, but now it effectively visualizes motion.

Step 5: Risk Register

Item	P	I	Risk (P*I)	Recognized	Mitigated/ Resolved	ROAM	How
Had another midterm to study for	2	40	100	24-Mar-25	Mitigated	M	Pushed the final project tast to another day
Had a medical procedure	3	20	60	14-Mar-25	Resolved	O	Test it. If mine not working, contact instructor
Board was not turning on	3	40	120	30-Mar-25	Resolved	O	Ended up getting another one from a student that graduated
			0				
			0				
			0				
			0				
			0				
			0